

Kaifa Lu-CV

Kai-Fa Lu

Post-Doctoral Research Associate in Civil & Construction Engineering
Center of Excellence in Capacity-building for REsilient Housing (CECREH)
Texas Tech University, Lubbock, TX, USA

Phone: +1-(352)-871-4546

Email: kflu.917@gmail.com

Website: https://kaifalu.github.io/kaifalu_page/

Research Interests: Urban Resilience; Urban Planning AI; Traffic Modeling & Planning; Environmental Planning & Engineering; GeoAI & Big Data; Machine Learning;



EDUCATION

2021-2025: Ph.D. Degree in Urban and Regional Planning, University of Florida, FL, USA

- GPA: 3.90/4.00
- Advisor/Committee Chair: Prof. Zhong-Ren Peng
- Committee Member: Dr. Ruth L. Steiner (Transportation Planning), Dr. Jose C. Principe (Electrical and Computer Engineering), Dr. Shenhao Wang (Urban AI)
- Dissertation: *AI-Driven Approach to Optimize and Automate Shared Micromobility System Planning and Operation*

2021-2024: Master's Degree in Electrical and Computer Engineering, University of Florida, FL, USA

- GPA: 3.89/4.00
- Key Courses: *Machine Learning, Neural Networks and Deep Learning, Data Analytics and Decision Sciences, Time Series, Pattern Recognition, Computer Communications*

2018-2021: Master's Degree in Transportation Engineering, Shanghai Jiao Tong University, Shanghai, China

- GPA: 3.83/4.00
- Thesis: *Characterizing Traffic-related Pollutant Distribution Patterns Under the Impacts of Urban Viaduct and Street Canyon*

2014-2018: Bachelor's Degree in Transportation Engineering, Central South University, Changsha, China

- GPA: 92.40/100.00
- Thesis: *Optimization of Vehicle Routing Problem over Local Road Network within Changsha South Railway Station*

CERTIFICATES

2024: Machine Learning Certificate, Electrical and Computer Engineering, University of Florida, FL, USA

2024: Urban Analytics (URP) Certificate, Urban and Regional Planning, University of Florida, FL, USA

2024: Generative AI with Diffusion Models, NVIDIA, Inc., USA

2023: Data Parallelism: How to Train Deep Learning Models on Multiple GPUs, NVIDIA, Inc., USA

2022&2023: Certificate of Outstanding Merit, University of Florida International Center (UFIC), University of Florida, FL, USA

JOURNAL PAPER PUBLICATIONS

- J1. Peng, Z., **Lu, K.**, Liu, Y., Hou, Q., Zhang, Q. (2026). Symbiotic Planning Theory: The CORE Framework for Human-AI Cocreation in Urban Planning. *Journal of the American Planning Association*, 1-16. <https://doi.org/10.1080/01944363.2026.2640038>
- J2. **Lu, K.**, Liu, Y., Peng, Z., Zhai, W. (2026). Characterizing Performance Resilience of Transportation Networks against Hurricane Events. *Applied Geography*, 186, 103820. <https://doi.org/10.1016/j.apgeog.2025.103820>
- J3. Popoola, T., Andrews, J., **Lu, K.**, Hayhoe, K., Nejat, A. (2026). Modeling NFIP-Insured Housing Losses from Texas and Florida Flood Disasters: A Comparative Analysis of Hurricane Harvey, Tax Day Flood, and Hurricane Irma. *Urban Climate*, 68, 103029. <https://doi.org/10.1016/j.uclim.2026.103029>
- J4. Yang, X., **Lu, K.***, Li, X. (2026). Assessing Vulnerability of Rural Buildings to Tornadoes and Their Relationships with Building Attributes and Surrounding Land Uses. *International Journal of Disaster Risk Reduction*, 137, 106097. <https://doi.org/10.1016/j.ijdrr.2026.106097>
- J5. Hou, Q., Hafiz, D., Peng, Z., **Lu, K.** (2026). From Perception to Adoption: A State-of-the-Art Review of Theoretical Approaches to Fully Autonomous Vehicle Acceptance and Transportation System Impacts. *Transport Policy*. In Press.
- J6. Yang, X. & **Lu, K.*** (2025). Analysis of Rural Housing Vulnerability to Windstorms Using Machine Learning Techniques: A Case Study of “6.23” Tornado in Yancheng, China (in Chinese). *Architectural Journal*, 32, 2-8.
- J7. **Lu, K.**, Liu, Y., Peng, Z. (2025). Assessing the Impacts of Transit Systems and Urban Street Features on Bike-Sharing Ridership: A Graph-based Spatiotemporal Analysis and Prediction Model. *Journal of Transport Geography*, 128, 104356. <https://doi.org/10.1016/j.jtrangeo.2025.104356>
- J8. **Lu, K.**, Liu, Y., Peng, Z. (2025). Unraveling Urban Bike-sharing Dynamics: Spatiotemporal Imbalances in Bike Rentals and Returns in Washington D.C. *Cities*, 162, 105967. <https://doi.org/10.1016/j.cities.2025.105967>
- J9. Peng, Z., **Lu, K.**, Liu, Y., & Zhai, W. (2024). The Pathway of Urban Planning AI: From Planning Support to Plan-Making. *Journal of Planning Education and Research*, 44(4), 2263-2279. <https://doi.org/10.1177/0739456X231180568>
- J10. Liu, Y., Peng, Z., Hou, Q., & **Lu, K.** (2024). The Role, Opportunities, and Challenges of Generative AI in Comprehensive Planning of American Small Towns – Using ChatGPT as an Example (in Chinese). *Journal of Urban Regional Planning*, 16(1), 215-228.
- J11. Liu, Y., **Lu, K.**, Peng, Z., & Zhai, W. (2024). Autonomous Shuttle Acceptance in an American Suburban

- Context: A Revealed Preference Study in Lake Nona, Florida. *Travel Behaviour and Society*, 37, 100865. <https://doi.org/10.1016/j.tbs.2024.100865>
- J12. Miao, C., Peng, Z., Cui, A., He, X., Chen, F., **Lu, K.**, et al. (2024). Quantifying and Predicting Air Quality on Different Road Types in Urban Environments Using Mobile Monitoring and Automated Machine Learning. *Atmospheric Pollution Research*, 15(3), 102015. <https://doi.org/10.1016/j.apr.2023.102015>
- J13. Jin, M., Gallagher, J., Li, X., **Lu, K.**, Peng, Z., & He, H. (2024). Characterizing the Distribution Pattern of Traffic-related Air Pollutants in Near-road Neighborhoods. *Environmental Monitoring and Assessment*, 196(8), 767. <https://doi.org/10.1007/s10661-024-12917-3>
- J14. Lu, D., He, H., Zhao, H., **Lu, K.**, Peng, Z., & Li, J. (2023). Quantifying Traffic-related Carbon Emissions on Elevated Roads through On-road Measurements. *Environmental Research*, 231, 116200. <https://doi.org/10.1016/j.envres.2023.116200>
- J15. Wang, H., Hu, Q., Huang, C., **Lu, K.**, He, H., & Peng, Z. (2023). Quantification of Gaseous and Particulate Emission Factors from a Cargo Ship on the Huangpu River. *Journal of Marine Science and Engineering*, 11(8), 1580. <https://doi.org/10.3390/jmse11081580>
- J16. **Lu, K.**, Peng, Z. (2023). Impacts of Viaduct and Geometry Configurations on the Distribution of Traffic-related Particulate Matter in Urban Street Canyon. *Science of the Total Environment*, 858, 159902. <https://doi.org/10.1016/j.scitotenv.2022.159902>
- J17. Zhu, X., **Lu, K.**, Peng, Z., et al. (2022). Spatiotemporal Variations of Carbon Dioxide (CO₂) at Urban Neighborhood Scale: Characterization of Distribution Patterns and Contributions of Emission Sources. *Sustainable Cities and Society*, 78, 103646. <https://doi.org/10.1016/j.scs.2021.103646>
- J18. **Lu, K.**, Wang, H., Li, X., et al. (2022). Assessing the Effects of Non-local Traffic Restriction Policy on Urban Air Quality. *Transport Policy*, 115, 62-74. <https://doi.org/10.1016/j.tranpol.2021.11.005>
- J19. Jia, Y., **Lu, K.**, Zheng, T., et al. (2021). Effects of Roadside Green Infrastructure on Particle Exposure: A Focus on Cyclists and Pedestrians on Pathways Between Urban Roads and Vegetative Barriers. *Atmospheric Pollution Research*, 12, 1-12. <https://doi.org/10.1016/j.apr.2021.01.017>
- J20. **Lu, K.**, He, H., Wang, H., et al. (2020). Characterizing Temporal and Vertical Distribution Patterns of Traffic-emitted Pollutants Near an Elevated Expressway in Urban Residential Areas. *Building and Environment*, 172, 106678. <https://doi.org/10.1016/j.buildenv.2020.106678>
- J21. Yang, X., Li, X., **Lu, K.**, & Peng, Z. (2023). Integrating Rural Livelihood Resilience and Sustainability for Post-disaster Community Relocation: A Theoretical Framework and Empirical Study. *Natural Hazards*, 116(2), 1775-1803. <https://doi.org/10.1007/s11069-022-05739-4>
- J22. Zhao, H., He, H., **Lu, K.**, et al. (2022). Measuring the Impact of an Exogenous Factor: An Exponential Smoothing Model of the Response of Shipping to COVID-19. *Transport Policy*, 118, 91-100. <https://doi.org/10.1016/j.tranpol.2022.01.015>
- J23. Zhu, X., He, H., **Lu, K.**, et al. (2022). Characterizing Carbon Emissions from China V and China VI Gasoline Vehicles Based on Portable Emission Measurement Systems. *Journal of Cleaner Production*,

- 378(10), 134458. <https://doi.org/10.1016/j.jclepro.2022.134458>
- J24. Zhao, H., He, H., **Lu, K.**, et al. (2022). Characterizing the Distribution Pattern of Submicron and Coarse Particles on High-density Container Truck Roads through Mobile Monitoring. *Atmospheric Pollution Research*, 13(10), 101561. <https://doi.org/10.1016/j.apr.2022.101561>
- J25. Wang, D., Wang, H., **Lu, K.**, et al. (2022). Regional Prediction of Ozone and Fine Particulate Matter Using Diffusion Convolutional Recurrent Neural Network. *International Journal of Environmental Research and Public Health*, 19(7), 3988. <https://doi.org/10.3390/ijerph19073988>
- J26. Cai, W., Wang, H., Wu, C., **Lu, K.**, et al. (2021). Characterizing the Interruption-Recovery Patterns of Urban Air Pollution under the COVID-19 Lockdown in China. *Building and Environment*, 205, 108231. <https://doi.org/10.1016/j.buildenv.2021.108231>
- J27. Wang, D., Wang, H., Li, C., **Lu, K.**, et al. (2020). Roadside Air Quality Forecasting in Shanghai with a Novel Sequence-to-Sequence Model. *International Journal of Environmental Research and Public Health*, 17(24), 9471. <https://doi.org/10.3390/ijerph17249471>

MANUSCRIPTS UNDER REVIEW

- J28. **Lu, K.**, Hou, Q., Zhang, Q., Liu, Y., Peng, Z. (2026). Planning Automation of Shared Micromobility System Design: A Multi-Agent Deep Reinforcement Learning Approach. *Nature Computational Science*. (Under Review)
- J29. Liu, Y., **Lu, K.**, Zhang, Q., et al. (2026). Are Current Streets Suitable for Autonomous Bus Transit? User Perception and Street View Imagery Analysis across 12 U.S. Sites. *Sustainable Cities and Society*. (Under Review)
- J30. **Lu, K.**, Hou, Q., Liu, Y., Peng, Z. (2026). Spatial Dynamics of Campus-Centric Micromobility: Shared Dockless Scooter Usage in University vs. Non-University Towns. *Transportation Research Part D: Transport and Environment*. (Under Review)
- J31. **Lu, K.**, Hou, Q., Liu, Y., Peng, Z. (2026). Leveraging Shared Micromobility to Boost Transit Accessibility and Ridership: Evidence from Florida Cities on Multimodal Integration. *Travel Behavior and Society*. (Under Review)
- J32. **Lu, K.**, Hou, Q., Peng, Z. (2026). Impact of Shared Micromobility System Sizes on Usage Patterns and Their Planning Implications: A Meta-Analysis of 52 U.S. Cities. *Journal of Planning Education and Research*. (Under Review)
- J33. Liu, Y., **Lu, K.**, Hou, Q., et al. (2026). Can Discounted Uber Rides Connecting to Transit Help Regain Bus Ridership? Evidence from Pinellas County, Florida. *Transport Policy*. (Under the Second Round of Review)
- J34. Zhang, Q., Peng, Z., Hou, Q., **Lu, K.**, Wang, D., Liu, Y. (2026). Human-AI Collaboration in Transit Network Design: A Data-Driven Planning Case from Tampa. *Journal of Planning Education and Research*. (Under Review)
- J35. **Lu, K.**, Hou, Q., Andrews, J., Nejat, A. (2026). State and Local Governmental Capacity in Disaster Recovery: Evidence from CDBG-DR. *Nature Climate Change*. (Under Review)

- J36. **Lu, K.**, Hou, Q., Andrews, J., Nejat, A. (2026). Measuring Governmental Capacity in Disaster Recovery: Evidence from Texas CDBG-DR. *International Journal of Disaster Risk Reduction*. (Under Review)
- J37. **Lu, K.**, Hou, Q., Popoola, T., Hayhoe, K., Andrews, J., Nejat, A. (2026). Projecting Future Climate-Housing Exposure through Integrated Climate, Urban Growth, and Social Vulnerability Assessment: A Harris County, Texas Case Study. *Urban Climate*. (Under Review)
- J38. **Lu, K.**, Hou, Q., Peng, Z. (2026). Spatiotemporal Multi-Graph Diffusion Learning for Predicting Transportation Network Performance and Climate Resilience. *The 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems, Riverside, CA, USA*. (Under Review)
- J39. **Lu, K.**, Hou, Q., Popoola, T., Hayhoe, K., Andrews, J., Nejat, A. (2026). Wildfire Risk Assessment in the United States: A State-of-the-Art Review of Synthesizing Contexts, Modeling Approaches, and Risk Indicators. *Risk Analysis*. (Under Review)
- J40. Andrews, J., **Lu, K.**, Popoola, T., Hayhoe, K., Nejat, A. (2026). Mapped Flood Zone Exposure and Flood Burden Recorded in Assistance Data: The Hydrologic Blind Spot of United States Flood Insurance. *Nature Communications*. (Under Review)
- J41. Andrews, J., **Lu, K.**, Popoola, T., Hayhoe, K., Nejat, A. (2026). Climate Extremes and Insurance Exposure: Projecting Future National Flood Insurance Program (NFIP) Claim Counts in Harris County, Texas. *Natural Hazards*. (Under Review)
- J42. Andrews, J., Nejat, A., **Lu, K.** (2026). Narrative Signals as Leading Indicators of Disaster Recovery Program Execution Context-Dependent Evidence from Texas CDBG-DR Grant Portfolios, 2009-2025. *International Journal of Disaster Risk Reduction*. (Under Review)

BOOK CHAPTERS

- B1. Peng, Z., **Lu, K.**, Jin, M., et al. (2022). *China's Metro Explosion: Lessons from China's Big Four Cities*. In: Landis, J. D. (eds) Megaprojects for Megacities. Edward Elgar Publishing, Inc., USA.
<http://dx.doi.org/10.4337/9781803920634>
- B2. Peng, Z., Zhai, W., **Lu, K.** (2022). *Smart, Sustainable, and Resilient Transportation System*. In: Li, B., Shi, X., Zhu, AX., Wang, C., Lin, H. (eds) New Thinking in GIScience. Springer, Singapore.
https://doi.org/10.1007/978-981-19-3816-0_34

TECHNICAL REPORTS

- R1. Peng, Z., **Lu, K.**, Liu, Y. (2024). *Micromobility Analytics in Florida: Usage Patterns, Public Transit Synergies, and Crash Insights*. Florida Department of Transportation.
<https://rosap.ntl.bts.gov/view/dot/86120>
- R2. Peng, Z., Liu, Y., Hou, Q., **Lu, K.** (2023). *A Synthesis of Economic Impact Assessment of Transit Services*. Florida Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/84601>
- R3. Peng, Z., **Lu, K.**, Liu, Y. (2023). *Examining Data Needs and Implementation Process of AV-Based Microtransit Service: A Case Study in Lake Nona*. Florida Department of Transportation.

TEACHING

University of Florida, Dept. Urban and Regional Planning, FL, USA

- Spring 2025 GIS Automation for Geospatial Modeling and Analysis (Graduate level) Teaching Assistant
- Fall 2024 Transportation Policy and Planning (Graduate level: Online & Hybrid) Teaching Assistant
- Spring 2024 Transportation and Land Use Modeling (Graduate level) Teaching Assistant
- Fall 2023 Planning for Climate Change (Graduate level) Teaching Assistant
- Spring 2022 Transportation and Land Use Modeling (Graduate level) Co-Instructor

Shanghai Jiao Tong University, Dept. Transportation Engineering, Shanghai, China

- Spring 2019 Operations Research (Undergraduate level) Teaching Assistant

ACADEMIC SERVICES

Mentor-Mentee Experience

- Texas Tech University (Ph.D. level): Hussein Orekoya, Ph.D. Dissertation 2025&2026
- Texas Tech University (Ph.D. level): Prashant Aryal, Ph.D. Dissertation 2025&2026
- Texas Tech University (Ph.D. level): Temidayo Popoola, Ph.D. Dissertation 2025&2026
- Texas Tech University (Ph.D. level): Amin Sobhani, Ph.D. Dissertation 2025&2026
- University of Florida (Ph.D. level): Khalid A. Aljuhani, Ph.D. Dissertation 2024&2025
- University of Florida (Ph.D. level): Qing Hou, Ph.D. Dissertation 2025
- University of Florida (Graduate level): Yue Dong, Master's Thesis 2023
- Shanghai Jiao Tong University (Graduate level): Xinghang Zhu, Master's Thesis 2020&2021

Journal Reviewers

- Journal of the American Planning Association 2025&2026
- Journal of Planning Education and Research 2025&2026
- Travel Behaviour and Society 2025&2026
- Journal of Environmental Management 2023
- Humanities and Social Sciences Communications 2024
- Scientific Reports 2024&2025
- Transportation Research Part D: Transport and Environment 2025
- Stochastic Environmental Research and Risk Assessment 2024

Affiliations

- Center of Excellence in Capacity-building for RESilient Housing (CECREH), Postdoctoral
- International Association for China Planning (IACP), Student Member
- International Center for Adaptation Planning and Design (iAdapt), Member
- University of Florida Transportation Institute (UFTI), Student Member

- The Committee on Extreme Weather and Climate Change Adaptation (AMR50), Friend Member

Conference Events

- Co-hosted “*AI and Cities: An International Forum for Innovation and Collaboration*” at the University of Florida, 2025
- Co-organized “*CECREH Inaugural Summit*” at Texas Tech University, 2025

AWARDS & SCHOLARSHIP

- **Best Presentation/Research Award**, AI and Cities: An International Forum for Innovation and Collaboration, hosted by the College of Design, Construction and Planning, University of Florida, USA, 2025
- **WRS Infrastructure & Environment Inc. Award in Memoriam of Mario Ripol for Outstanding Achievement in Planning Information and Analysis**, College of Design, Construction and Planning, University of Florida, USA, 2025
- **Merit Commendation** (PhD Research Poster), Graduate Student Research Symposium, College of Design, Construction and Planning, University of Florida, USA, 2023
- **COSCO maritime scholarship** (top 5%), China Ocean Shipping Company, China, 2021
- **Outstanding Graduates of Shanghai** (top 3%), Shanghai, China, 2021
- **The Second Prize** in the National Graduate Mathematical Contest in Modeling, China, 2019
- **Outstanding Undergraduates of Hunan Province** (top 3%), Changsha, Hunan Province, China, 2018
- **The Meritorious Winner** in the American Undergraduate Mathematical Contest in Modeling, USA, 2017
- **National Scholarship** (top 3%), The Ministry of Education of China, 2017
- **The First Prize** in the National Undergraduate Mathematical Contest in Modeling, China, 2016
- **The Third Prize** in the National Undergraduate Mathematical Contest, China, 2015
- **National Encouragement Scholarship** (top 5%), The Ministry of Education of China, 2015&2016
- **First-class Scholarship** (top 10%) of **Shanghai Jiao Tong University & Central South University**, China, 2015-2021

PRESENTATIONS & TALKS

Conference Presentations

- C1. **2026.06: 51st Annual Natural Hazards Workshop Research Meeting.** *State and Local Governmental Capacity in Disaster Recovery: Evidence from CDBG-DR.* Broomfield, Colorado, USA.
- C2. **2026.06: 51st Annual Natural Hazards Workshop Poster Session.** *Measuring Governmental Capacity in Disaster Recovery: Evidence from Texas CDBG-DR.* Broomfield, Colorado, USA.
- C3. **2025.10: Association of Collegiate Schools of Planning (ACSP) Conference.** *AI-Driven Approach to Optimize and Automate Shared Micromobility System Design and Planning.* Minneapolis, MN, USA.
- C4. **2025.01: Transportation Research Board 104th Annual Meeting.** *Characterizing Performance Resilience of Transportation Networks against Extreme Weather Events.* Washington, D.C., USA.
- C5. **2025.01: Transportation Research Board 104th Annual Meeting.** *Integrating Public Transit Effects*

- and Street View Imagery into a Dynamic Spatiotemporal Graph-based Machine Learning Model for Predicting Bike-sharing Ridership.* Washington, D.C., USA.
- C6. **2024.11: Association of Collegiate Schools of Planning (ACSP) Conference.** *Impact of Shared Micromobility System Sizes on Usage Patterns: Planning Implications.* Seattle, WA, USA.
- C7. **2024.07: The 18th International Association for China Planning (IACP) Conference.** *Impact of Shared Micromobility System Sizes on Usage Patterns and Their Planning Implications.* Hangzhou, China.
- C8. **2024.01: Transportation Research Board 103rd Annual Meeting.** *Demystifying the Spatiotemporal Heterogeneity of Rental-Return Imbalance on Bike-Sharing Systems: A Bayesian Additive Regression Trees (BART) Model.* Washington, D.C., USA.
- C9. **2023.10: Association of Collegiate Schools of Planning (ACSP) Conference.** *Leveraging Deep Learning with Geospatial Data Analytics for Quantification and Prediction of Performance Resilience of Transportation Networks against Extreme Weather Events.* Chicago, IL, USA.
- C10. **2023.07: The 17th International Association for China Planning (IACP) Conference.** *Using Origin-Destination Flow Graph and Public Transit Information to Enhance Short-Term Ridership Prediction in Bike-Sharing Systems.* Tianjin, China.
- C11. **2023.01: Transportation Research Board 102nd Annual Meeting.** *Characterization and Prediction of Transportation Network Resilience: A Spatiotemporal Graph Diffusion Convolutional Recurrent Neural Network Approach.* Washington, D.C., USA.
- C12. **2023.01: Transportation Research Board 102nd Annual Meeting.** *Characterizing Carbon Emissions from China V and China VI Gasoline Vehicles Based on Portable Emission Measurement Systems.* Washington, D.C., USA.
- C13. **2023.01: Transportation Research Board 102nd Annual Meeting.** *Characterizing the Traffic-related Carbon Emission Factors on Elevated Roads Based on On-road Measurements.* Washington, D.C., USA.
- C14. **2021.01: Transportation Research Board 100th Annual Meeting.** *Investigating Pedestrians' Exposure to Traffic-Related PM and BC at Intersections: A Case Study in Shanghai, China.* Washington, D.C., USA.
- C15. **2020.01: Transportation Research Board 99th Annual Meeting.** *Characterization of Traffic-related Pollutant Distribution Patterns in Urban Residential Areas with an Elevated Expressway.* Washington, D.C., USA.

Invited Talks

- T1. **2024.06: Guest Lecture at Chang'an University.** *Integrating Public Transit Effects and Street View Imagery into a Dynamic Spatiotemporal Graph-Based Machine Learning Model for Analyzing and Predicting Bike-Sharing Ridership.* Xi'an, China.

PROJECT EXPERIENCES

Research Lab, Center of Excellence in Capacity-building for REsilient Housing, 2025.10-now, Postdoctoral Research Associate

- Funded by the U.S. Department of Housing and Urban Development (HUD) and National Science Foundation (NSF), \$2M
- Develop GIS-based dashboards that facilitate data management and querying of disaster recovery programs and housing data
- Apply SEM, NLP, and LLMs to analyze narratives and numeric data of disaster recovery program performance to improve state and local governmental capacity for administering HUD funds
- Assess climate risks for underserved communities, build capacity for equitable disaster recovery, and develop tools and simulations to inform climate-resilient policies

Ph.D. Dissertation Research, AI-Driven Approach to Optimize and Automate Shared Micromobility System Planning and Operation, 2023.08-2025.08, Principal Investigator

- Apply spatial data analysis and machine learning (i.e., Bayesian Additive Regression Tree) models to the data of bike/scooter trips and influencing factors to reveal usage patterns and underlying causes
- Use deep learning models (i.e., Graph Neural Networks) to predict bike- or scooter-sharing ridership
- Develop a multi-agent deep reinforcement learning approach to optimize and automate shared micromobility system design toward usage, equity, cost-effectiveness, and multimodal integration

Research Project, Assessment of Modelling Framework for Micromobility, 2023.03-2024.10, Research Assistant

- Funded by the Florida Department of Transportation (FDOT), FL, USA, \$300k
- Develop a modeling framework for micromobility analytics that integrates survey analysis, spatial data analysis, and machine learning to understand their usage patterns, mode choices, crash patterns, and relationships with public transit in Florida
- Micromobility analytics in Florida: Usage patterns, public transit synergies, and crash insights

Research Project, A Synthesis of Economic Impact Assessment of Transit Services, 2023.06-2023.12, Research Assistant

- Funded by the Florida Department of Transportation (FDOT), FL, USA, \$150k
- Examine the application of Benefit-Cost Analysis (BCA) and Economic Impact Analysis (EIA) methodologies in public transit systems and highlight geographical variations and trends
- Analyze themes, trends, and gaps in existing research through extensive literature review, bibliometric mapping, and case studies from key metropolitan areas

Research Project, Florida's Transportation Revenue Forecasting and Allocation Process and Modeling – Phase I, 2022.07-2022.10, Research Assistant

- Funded by the Florida Department of Transportation (FDOT), FL, USA, \$200k

- Develop models to understand the forces that will affect future transportation revenues in Florida
- Delineate the whole process and concerns of allocating the revenue to Districts and MPOs

Research Project, *Examining Data Needs and Implementation Process of AV-based Microtransit Service: A Case Study in Lake Nona*, 2021.06-2023.01, Research Assistant

- Funded by the Florida Department of Transportation (FDOT), FL, USA, \$350k
- Develop a framework for examining various aspects of the AV-based microtransit system, including policy and government support, infrastructure and technology, service and management, financial sustainability, ridership, and community impact.

Research Project, *Microtransit and Micromobility Inventory in the State of Florida*, 2021.07-2022.03, Research Assistant

- Funded by the Florida Department of Transportation (FDOT), FL, USA, \$280k
- Inventory all forms of microtransit and micromobility in Florida, map the geofenced service areas, and identify their relationship, if any, with transit agencies

Research Project, *Study on the Impact of Urban Transportation Policy and Facility Construction on Atmospheric Environment*, 2018.09-2021.03, Research Assistant, \$150k

- Funded by the National Planning Office of Philosophy and Social Science, Beijing, China
- Investigate the potential of different transportation policies (i.e., traffic restriction policy) and infrastructure planning (i.e., viaduct, street canyon) in improving urban air quality, as well as their impacts on the spatiotemporal distribution patterns of traffic-related pollutants

Research Project, *Vertical Observation Technologies of Atmospheric Pollution Based on Unmanned Aerial Vehicle and Heavy Load Airship*, 2018.09-2021.03, Research Assistant

- Funded by the Ministry of Science and Technology, Beijing, China, \$750k
- Patent: “He, H., Li, B., Cao, R., Lu, K., & Luo, Z. (2022). *An Intelligent Pod System for Three-Dimensional Atmospheric Environment Monitoring* (China Patent, No. CN215932395U). China National Intellectual Property Administration: Patent and Trademark Office.”

Research Project, *RFID-based Automatic Toll System for Roadside Parking*, 2016.06-2017.07, Principal Investigator

- Funded by the Hunan Education Commission, Changsha, Hunan Province, China, \$3k
- Develop an RFID-based automatic toll system for roadside parking to monitor illegal parking behaviors and improve charging efficiency and accuracy

Design & Planning Practice Project, *China Railway Guangzhou & Wuhan Groups*, 2017-2018, Transportation Planner & Engineer Internship

- Cognition practice of passenger and freight transport in Zhuzhou railway station, 2017.04-2017.05

- Production practice of Centralized Traffic Control for train scheduling in Wuhan & Guangzhou Dispatch Centers, 2018.10-2018.11
- Railway station design and planning for freight transport, 2017.09-2018.01

EXPERTISE & TECHNICAL SKILLS

Programming Language: Python, MATLAB, R, JAVA, C (++)

Software: ArcGIS, AutoCAD, PTV VISUM, TransCAD, SketchUp, FLUENT, MS OFFICE, ORIGIN